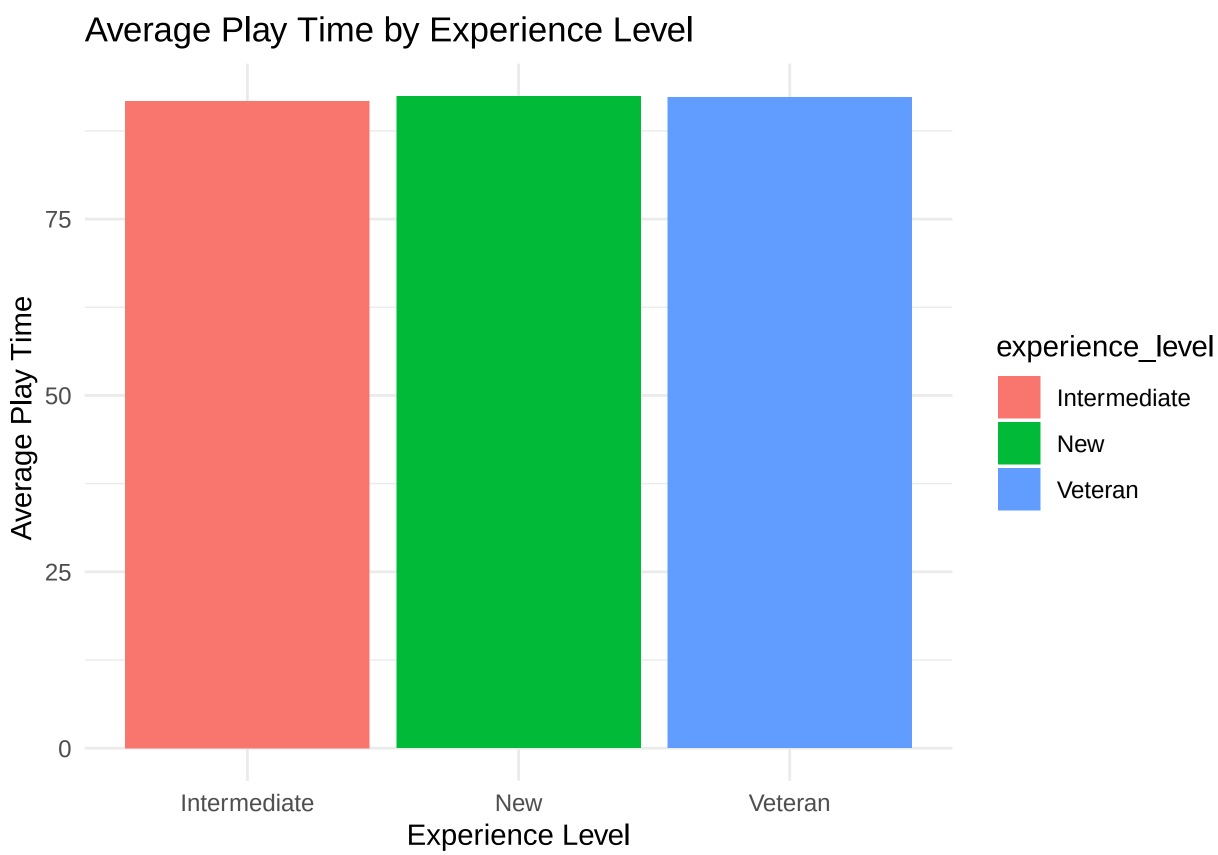
```r
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```r
library(ggplot2)
library(knitr)
setwd("/Users/aka6208/Desktop/game assignment")

#Question 1

players <- read.csv("players.csv")
sessions <- read.csv("sessions.csv")
games <- read.csv("games.csv")

players$signup_date <- as.Date(players$signup_date)
players <- players %>%
mutate(
  experience_level = case_when(
    format(signup_date, "%Y") <"2023"~ "Veteran",
    format(signup_date, "%Y") == "2023"~ "Intermediate",
    format(signup_date, "%Y") > "2023"~ "New"
  )
)
merged_data <- sessions %>%
  left_join(players, by = "player_id")

summary_table <- merged_data %>%
  group_by(experience_level) %>%
  summarise(
    number_players = n_distinct(player_id),
    average_play_time = mean (play_time_minutes, na.rm =TRUE),
    average_score = mean (score, na.rm = TRUE)
  )
kable(summary_table)
```

experience_level	number_players	average_play_time	average_score
Intermediate	1825	91.72935	498.2044
New	1340	92.41463	495.9630
Veteran	1764	92.26217	501.0407

```r
ggplot(summary_table,
       aes(
         x = experience_level,
           y = average_play_time,
           fill = experience_level
         )) +
         geom_bar(stat = "identity") +
         labs(
           title = "Average Play Time by Experience Level",
           x = "Experience Level",
           y = " Average Play Time"
         ) +
         theme_minimal()
```

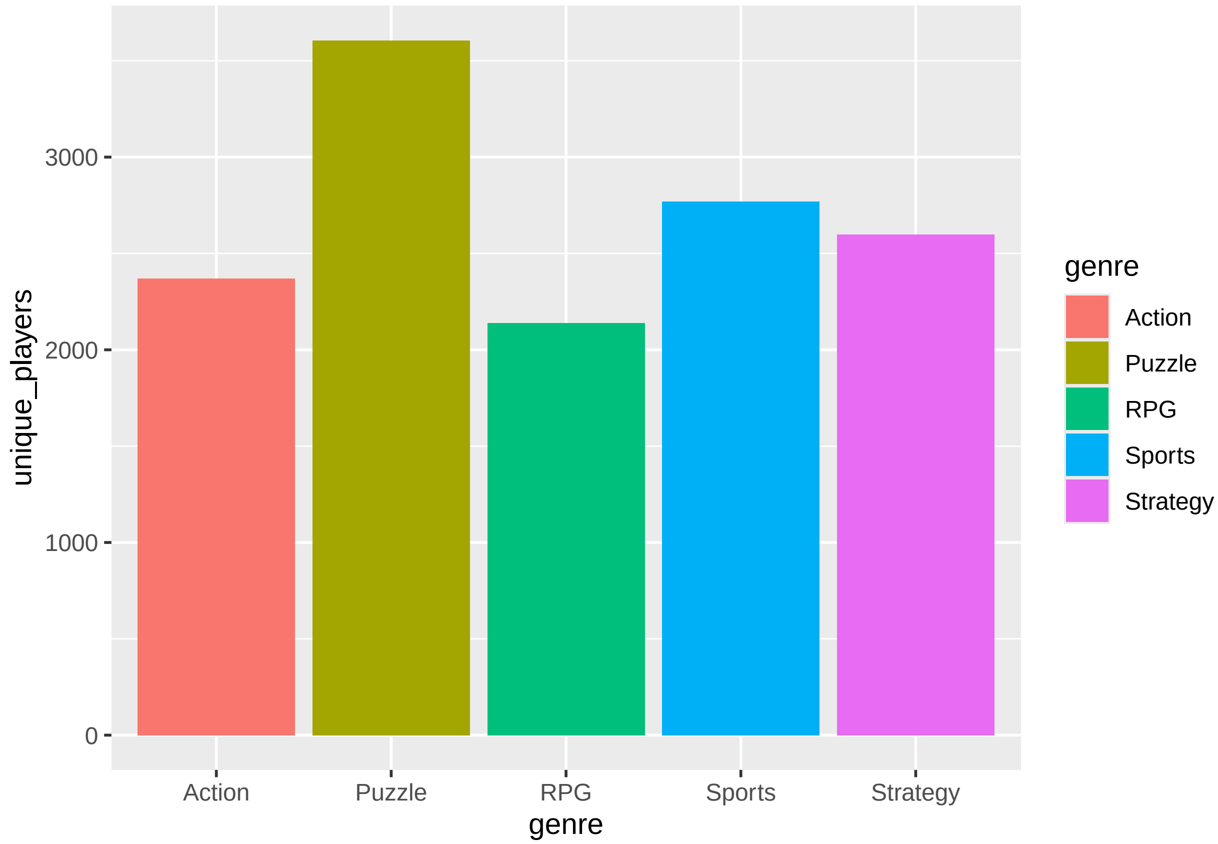



```r
#Question 2

# Take the sessions dataframe
# then join the games dataframe
# using the common column game_id
  genre_data <- sessions %>%
    left_join (games, by = "game_id")

#take the joined data
#then group bu the data by genre
  genre_summary <- genre_data %>%
```

```
  group_by (genre) %>%

 summarise(

      average_play_time = mean(play_time_minutes, na.rm = TRUE),
      average_score = mean(score, na.rm= TRUE),
      number_session = n(),
      unique_players = n_distinct(player_id)
    )
   kable(genre_summary)
```

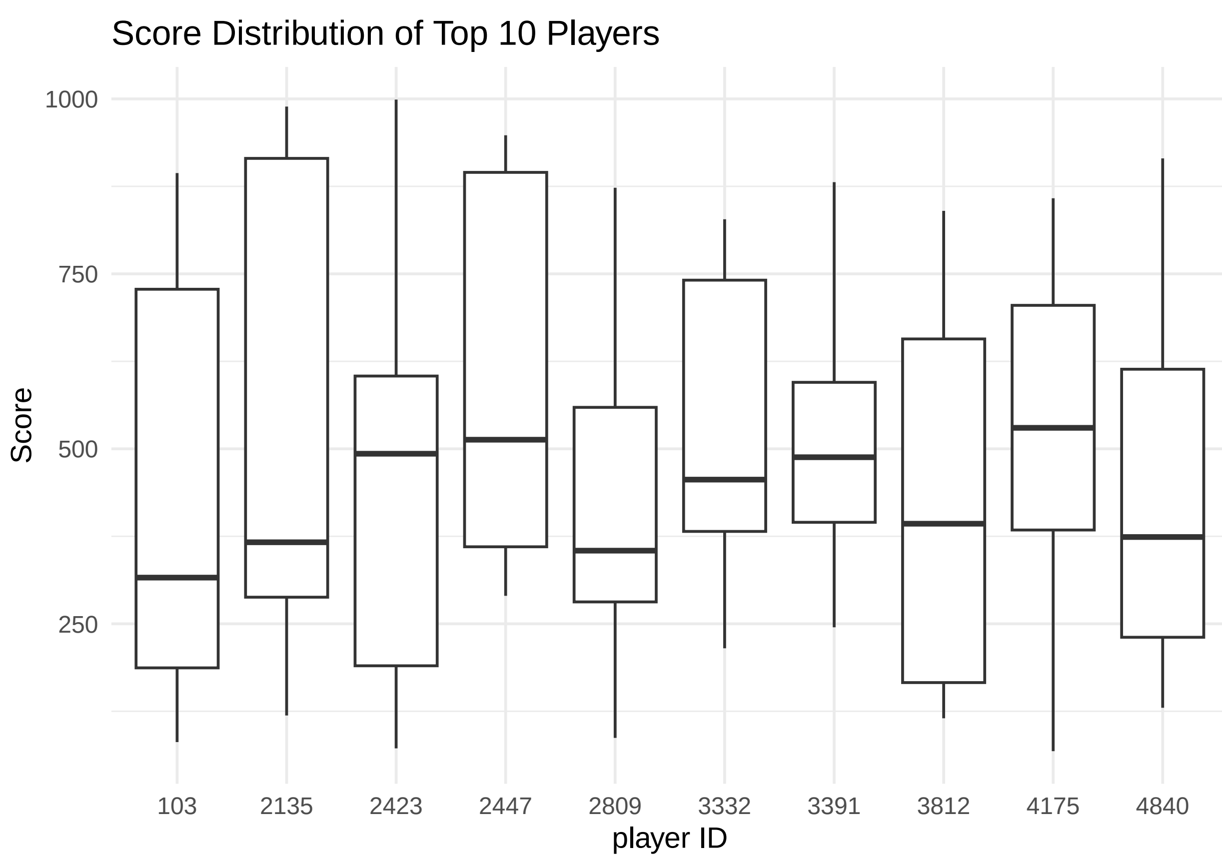
genre	average_play_time	average_score	number_session	unique_players
Action	92.09816	493.0569	3199	2370
Puzzle	91.67226	496.5239	6371	3606
RPG	91.41153	499.6938	2792	2139
Sports	92.58573	503.3790	3995	2769
Strategy	92.88334	500.8435	3643	2599

```
  ggplot(
    genre_summary,
    aes(
      x=genre,
      y=unique_players,
      fill = genre
    )
  )+
    geom_bar(stat = "identity"
    )
```




```
#Question 3

top_players <- sessions %>%
# Take the sessions dataframe
group_by(player_id) %>%
#Then group the data by player_id

summarise(
 total_play_time = sum(play_time_minutes, na.rm = TRUE)
) %>%
#create a summary table and calculate total play time for each player

arrange(desc(total_play_time)) %>%
# arrange players from highest to lowest play time

slice_head (n=10)
# Select the top 10 players only

top_player_data <- sessions %>%
#Take the sessions dataframe


filter(player_id %in% top_players$player_id)

top_summary <- top_player_data %>%
group_by(player_id) %>%
```

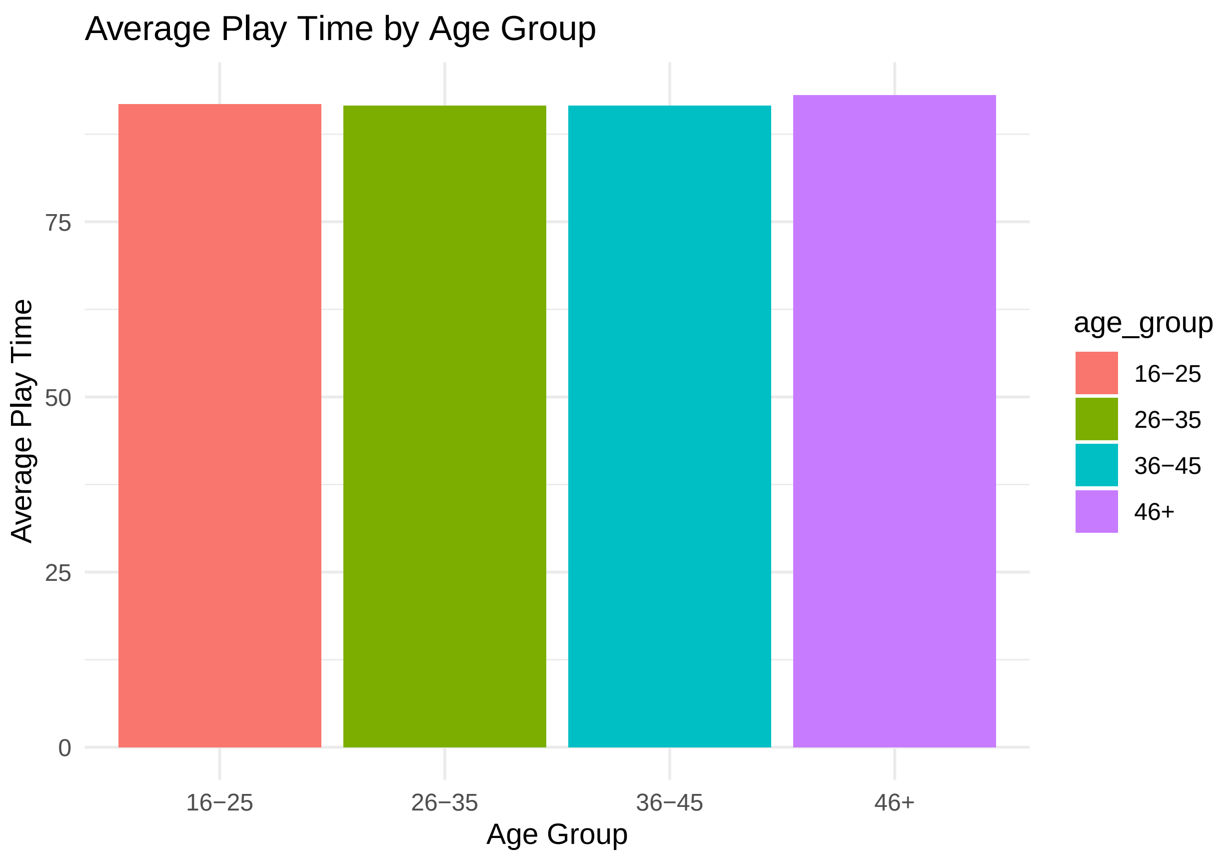
```r
#create summary statistics

summarise(
total_sessions = n(),

average_play_time = mean(play_time_minutes, na.rm = TRUE),

average_score = mean (score, na.rm=TRUE)
)
ggplot(
top_player_data,
aes(
x=factor(player_id),
y=score
)
) +
geom_boxplot()+
labs(
title = "Score Distribution of Top 10 Players",
    x="player ID",
    y="Score"
) +

theme_minimal()
```



```r
#Question 4
players <- players %>%
  mutate(
    age_group = case_when(
      age >= 16 & age <=25 ~"16-25",
      age>= 26 & age <= 35 ~ "26-35",
      age >= 36 & age <= 45 ~"36-45",
      age >=46 ~"46+"
    )
  )

age_data <- sessions %>%
  left_join(players, by="player_id")
#join sessions data with players data
age_summary <- age_data %>%
  group_by(age_group) %>%
  summarise(
    average_play_time=mean(play_time_minutes, na.rm =TRUE),
    average_score=mean(score, na.rm = TRUE),
    number_sessions = n()
  )

kable(age_summary)
```

age_group	average_play_time	average_score	number_sessions
16-25	91.79946	496.6914	4413
26-35	91.56831	503.5342	4538
36-45	91.57548	498.0227	4756
46+	93.11298	496.7149	6293

```r
ggplot(age_summary,
       aes(
         x=age_group,
         y=average_play_time,
         fill = age_group
       )
       )+
  geom_bar(stat = "identity") +
  labs(
    title= "Average Play Time by Age Group",
    x="Age Group",
    y="Average Play Time"
  )+
  theme_minimal()
```


Average Play Time by Age Group
Average Play Time
75
50
25
0
16−25
26−35
36−45
46+
Age Group
age_group
16−25
26−35
36−45
46+